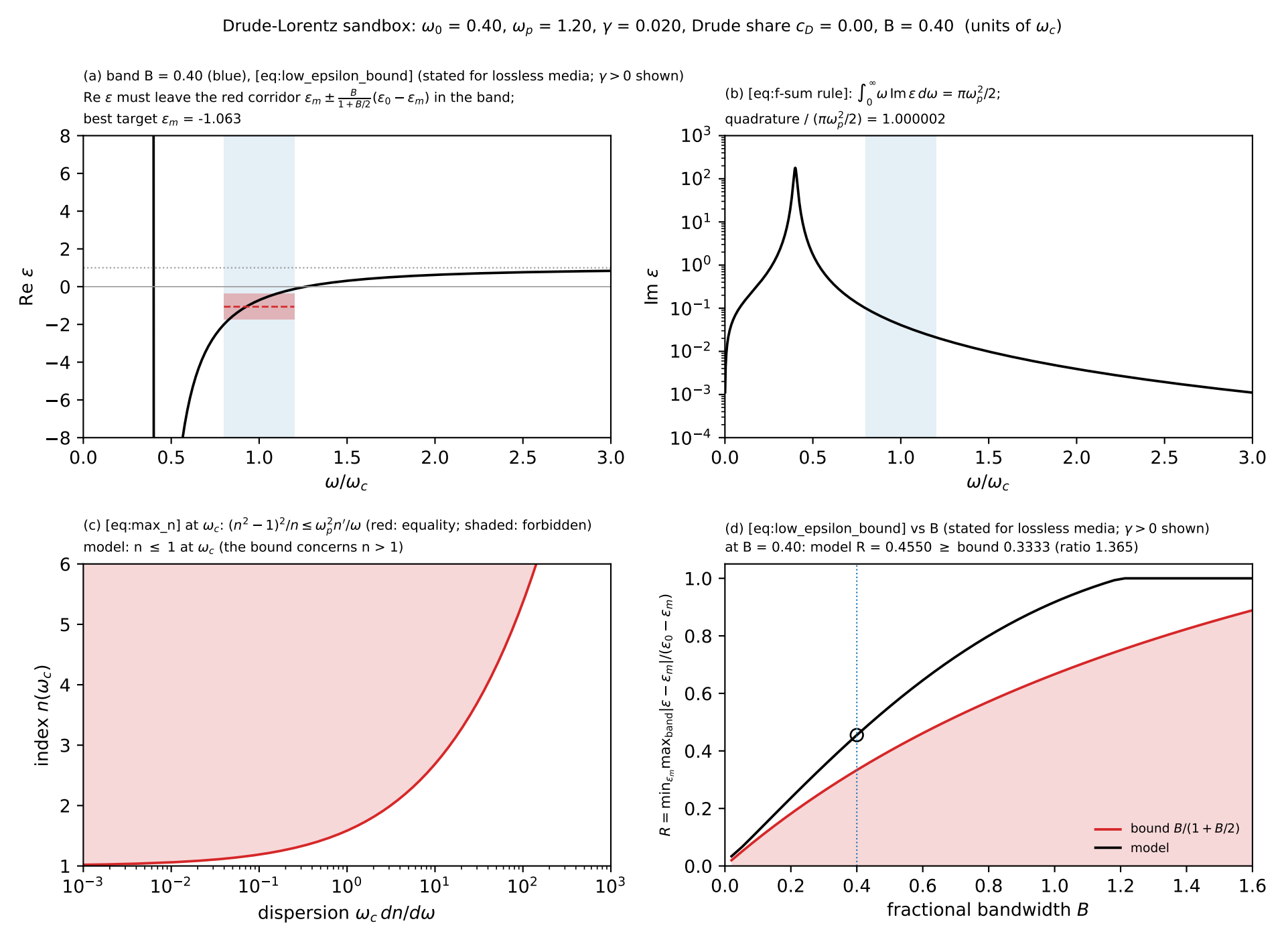
Drude-Lorentz sandbox: ω_0 = 0.40, ω_p = 1.20, γ = 0.020, Drude share c_D = 0.00, B = 0.40 (units of ω_c)
(a) band B = 0.40 (blue), [eq:low_epsilon_bound] (stated for lossless media; $\gamma > 0$ shown)
Re ε must leave the red corridor $\varepsilon_m \pm \frac{B}{1+B/2}(\varepsilon_0 - \varepsilon_m)$ in the band;
best target ε_m = -1.063
Re ε
ω/ω_c
(b) [eq:f-sum rule]: $\int_0^\infty \omega \,\mathrm{Im}\,\varepsilon \, d\omega = \pi\omega_p^2/2$;
quadrature / $(\pi\omega_p^2/2)$ = 1.000002
Im ε
ω/ω_c
(c) [eq:max_n] at ω_c: $(n^2-1)^2/n \le \omega_p^2 n'/\omega$ (red: equality; shaded: forbidden)
model: n ≤ 1 at ω_c (the bound concerns n > 1)
index $n(\omega_c)$
dispersion $\omega_c\, dn/d\omega$
(d) [eq:low_epsilon_bound] vs B (stated for lossless media; $\gamma > 0$ shown)
at B = 0.40: model R = 0.4550 ≥ bound 0.3333 (ratio 1.365)
$R = \min_{\varepsilon_m} \max_{\text{band}} |\varepsilon - \varepsilon_m| / (\varepsilon_0 - \varepsilon_m)$
bound $B/(1+B/2)$
model
fractional bandwidth B